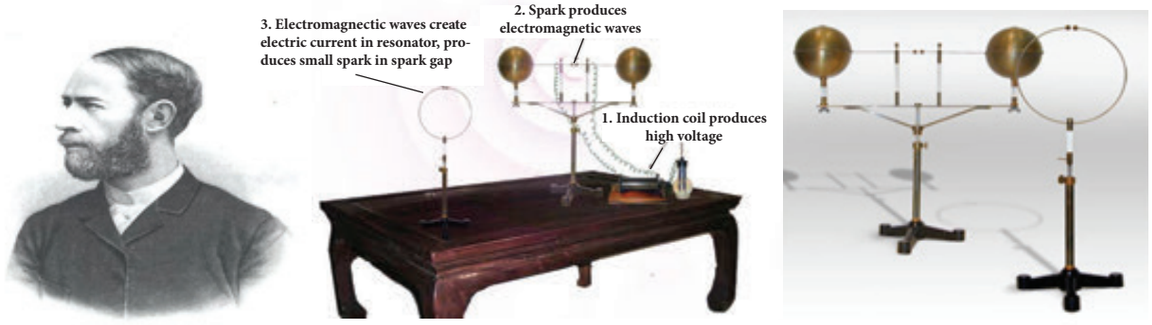


Figure 5.7 Hertz experiment (a) Heinrich Rudolf Hertz (1857 – 1894) (b) Hertz apparatus

coil is maintained at very high potential, air between the electrodes gets ionized and spark (spark means discharge of electricity) is produced. The gap between electrode (ring type – not completely closed and has a small gap in between) kept at a distance also gets spark. This implies that the energy is transmitted from electrode to the receiver (ring electrode) as a wave, known as electromagnetic waves. If the receiver is rotated by 90° - then no spark is observed by the receiver. This confirms that electromagnetic waves are transverse waves as predicted by Maxwell. Hertz detected radio waves and also computed the speed of radio waves which is equal to the speed of light (3×10^8 m s^{-1}).

Properties of electromagnetic waves

1. Electromagnetic waves are produced by any accelerated charge.
2. Electromagnetic waves do not require any medium for propagation. So electromagnetic wave is a non-mechanical wave.
3. Electromagnetic waves are transverse in nature. This means that the oscillating electric field vector, oscillating magnetic field vector and propagation vector (gives direction of propagation) are mutually perpendicular to each other. The electric and magnetic fields are in the y and z directions respectively and the direction of propagation is along x direction. This is shown in Figure 5.8.
4. Electromagnetic waves travel with speed which is equal to the speed of light in vacuum or free space, $c = \frac{1}{\sqrt{\varepsilon_\circ \mu_\circ}} = 3\times10^8 \,\mathrm{m\,s^{-1}}$, where ε_o is the permittivity of free space or vacuum and μ_o is the permeability of free space or vacuum (refer Unit 1 for permittivity and Unit 3 for permeability).
5. In a medium with permittivity ε and permeability μ, the speed of electromagnetic wave is less than speed in free space or vacuum, that is, $v < c$. In a medium of refractive index, $\mu = \frac{c}{v} = \frac{\frac{1}{\sqrt{\varepsilon_\circ \mu_\circ}}}{\frac{1}{\sqrt{\varepsilon\mu}}} \Rightarrow \mu = \sqrt{\varepsilon_r \mu_r}$, where ε_r is the relative permittivity of the medium (also known as dielectric constant) and μ_r is the relative permeability of the medium.
6. Electromagnetic waves are not deflected by electric field or magnetic field.

UNIT-5(XII-Physics_Vol-1).indd 289 04-03-2019 11:05:43